

Problem Set 3: Quantile Regression

Econ 692

Deadline: Thursday, March 20. Feel free to do computer exercises in any package you like (not necessarily R) and let me know if you need your data in some other format.

Rules: For full credit, please submit your computer code (script) with added comments about what each line of your code does (similar to the lab example posted on the BB). Two or more identical codes or solutions will receive a grade of zero each (please see the syllabus).

Assignment:

For this assignment, you'll be using a sub-sample of 2015 Natality Data from the National Center for Health Statistics (**Bwght.RData**). This data set contains the following data on newborn infants:

- *Married* = 1 if a baby's mother is married, = 0 otherwise;
- *PrenatalCare* = 1 if a baby's mother received prenatal care during the first trimester, = 0 otherwise;
- *Boy* = 1 if a baby is a boy, = 0 otherwise;
- *Weight* = weight of a baby, in grams.

A policymaker is thinking about benefits of making prenatal care more accessible. So to help that policymaker, your job is to analyze the effect of prenatal care on a baby's birth weight.

(a) (5 pts) Set up a regression model that allows you to evaluate the effect of a mother's receiving prenatal care on her baby's birth weight:

- What is your dependent variable?
- What is your key explanatory variable?
- What other variables would you include in this model, and why?
- Do you think that your key explanatory variable is exogenous or endogenous in your model? Why?

- (b) (10 pts) Estimate your model by OLS. Give economic interpretations to all coefficient estimates (including the intercept).
- (c) (5 pts) Discuss which standard errors you should use in part (b).
- (d) (20 pts) Fit a quantile regression for $\tau = (0.10, 0.25, 0.50, 0.75, 0.90)$ with same explanatory variables as in (b). Use “tau=c(0.10,0.25,0.5,0.75,0.90)” (sans quotation marks) in the **rq** function to get this specific set of quantiles.
- (e) (10 pts) Test the null hypothesis that the effect of prenatal care is homogeneous. Use 5% significance level. What is your conclusion?
- (f) (10 pts) Plots quantile intercept and slope coefficients against τ . How do the coefficients change as τ increases?
- (g) (10 pts) What do the quantile coefficient plots tell you about the quantile effect of prenatal care on the conditional distribution of birth weights?
- (h) (10 pts) What does the quantile coefficient plot for the intercept represents in your model? Specifically: the distribution of birth weights for which sub-population of babies?
- (i) (10 pts) Use the graph from part (h) to plot the **cumulative distribution function** of birth weights for that sub-population. On the same graph, use your quantile regression results to plot the **cumulative distribution function** of birth weights that sub-population would have if they were to receive pre-natal care. We did a similar exercise in our lab class.
- (j) (10 pts) Discuss economic (policy) implications of your findings, with a focus on prenatal care.

Work to Submit on Blackboard:

1. An R script that I can run to reproduce your results. Add comments to each line of your script that describe what that line does. Missing comments will result in losing 10 points. If your script produces error messages, I will deduct 5 points for each error message I had to fix to make your script running (my starting point is loading the data). Your script should include the following parts: (b), (e), (d), (f).
2. A PDF file with answers to all other parts. You can submit your CDF picture either as a part of your PDF report or as a separate file (you can draw it by hand if you wish so as long as your picture faithfully reproduces your quantile regression estimates).